13. At 400 K, the root mean square (rms) speed of a gas **X** (molecular weight = 40) is equal to the most probable speed of gas **Y** at 60 K. The molecular weight of the gas **Y** is

Answer 0 1 2 3 **4** 5 6 7 8 9

14. The dissociation constant of a substituted benzoic acid at 25°C is 1.0×10^{-4}. The pH of a 0.01 M solution of its sodium salt is

Answer 0 1 2 3 4 5 6 7 **8** 9

15. The total number of α and β particles emitted in the nuclear reaction $^{238}_{92}U \rightarrow {}^{214}_{82}Pb$ is

Answer 0 1 2 3 4 5 6 7 **8** 9

16. The oxidation number of Mn in the product of alkaline oxidative fusion of MnO_2 is

Answer 0 1 2 3 4 5 **6** 7 8 9

17. The number of water molecule(s) directly bonded to the metal centre in $CuSO_4 \cdot 5H_2O$ is

Answer 0 1 2 3 **4** 5 6 7 8 9

18. The coordination number of Al in the crystalline state of $AlCl_3$ is

Answer 0 1 2 3 4 5 **6** 7 8 9

19. The total number of cyclic structural as well as stereo isomers possible for a compound with the molecular formula C_5H_{10} is

Answer 0 1 2 3 4 5 6 **7** 8 9

PART II : MATHEMATICS

SECTION – I

Single Correct Choice Type

This section contains 4 multiple choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct.

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20. If the sum of first n terms of an A.P. is cn^2, then the sum of squares of these n terms is

(A) $\frac{n(4n^2-1)c^2}{6}$ (B) $\frac{n(4n^2+1)c^2}{3}$

(C) $\frac{n(4n^2-1)c^2}{3}$ (D) $\frac{n(4n^2+1)c^2}{6}$

Answer ○ ○ ● ○

(A) (B) (C) (D)

21. A line with positive direction cosines passes through the point $P(2, -1, 2)$ and makes equal angles with the coordinate axes. The line meets the plane

$$2x + y + z = 9$$

at point Q. The length of the line segment PQ equals

(A) 1 (B) $\sqrt{2}$ (C) $\sqrt{3}$ (D) 2

Answer ○ ○ ● ○

(A) (B) (C) (D)

22. The normal at a point P on the ellipse $x^2 + 4y^2 = 16$ meets the x-axis at Q. If M is the mid point of the line segment PQ, then the locus of M intersects the latus rectums of the given ellipse at the points

(A) $\left(\pm\frac{3\sqrt{5}}{2}, \pm\frac{2}{7}\right)$ (B) $\left(\pm\frac{3\sqrt{5}}{2}, \pm\frac{\sqrt{19}}{4}\right)$

(C) $\left(\pm 2\sqrt{3}, \pm\frac{1}{7}\right)$ (D) $\left(\pm 2\sqrt{3}, \pm\frac{4\sqrt{3}}{7}\right)$

Answer ○ ○ ● ○

(A) (B) (C) (D)

23. The locus of the orthocentre of the triangle formed by the lines

$$(1 + p)x - py + p(1 + p) = 0,$$
$$(1 + q)x - qy + q(1 + q) = 0,$$

and $y = 0$, where $p \neq q$, is

(A) a hyperbola (B) a parabola

(C) an ellipse (D) a straight line

Answer ○ ○ ○ ●

(A) (B) (C) (D)

SECTION – II

Multiple Correct Choice Type

This section contains 5 multiple choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE** is/are correct.

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24. If

$$I_n = \int_{-\pi}^{\pi} \frac{\sin nx}{(1 + \pi^x)\sin x}\, dx, \quad n = 0, 1, 2, \ldots,$$

then

(A) $I_n = I_{n+2}$ (B) $\sum_{m=1}^{10} I_{2m+1} = 10\pi$

(C) $\sum_{m=1}^{10} I_{2m} = 0$ (D) $I_n = I_{n+1}$

Answer ● ● ● ○

(A) (B) (C) (D)

25. An ellipse intersects the hyperbola $2x^2 - 2y^2 = 1$ orthogonally. The eccentricity of the ellipse is reciprocal of that of the hyperbola. If the axes of the ellipse are along the coordinate axes, then

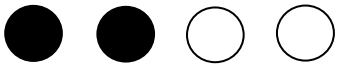
(A) Equation of ellipse is $x^2 + 2y^2 = 2$

(B) The foci of ellipse are $(\pm 1, 0)$

(C) Equation of ellipse is $x^2 + 2y^2 = 4$

(D) The foci of ellipse are $(\pm\sqrt{2}, 0)$

Answer



(A) (B) (C) (D)

26. For the function

$$f(x) = x\cos\frac{1}{x}, \quad x \geq 1,$$

(A) for at least one x in the interval $[1, \infty)$, $f(x+2) - f(x) < 2$

(B) $\lim\limits_{x\to\infty} f'(x) = 1$

(C) for all x in the interval $[1, \infty)$, $f(x+2) - f(x) > 2$

(D) $f'(x)$ is strictly decreasing in the interval $[1, \infty)$

Answer



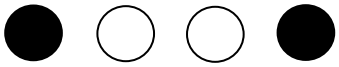
(A) (B) (C) (D)

27. The tangent PT and the normal PN to the parabola $y^2 = 4ax$ at a point P on it meet its axis at points T and N, respectively. The locus of the centroid of the triangle PTN is a parabola whose

(A) vertex is $\left(\frac{2a}{3}, 0\right)$ (B) directrix is $x = 0$

(C) latus rectum is $\frac{2a}{3}$ (D) focus is $(a, 0)$

Answer



(A) (B) (C) (D)

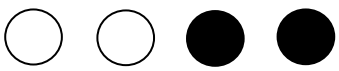
28. For $0 < \theta < \frac{\pi}{2}$, the solution(s) of

$$\sum_{m=1}^{6} \operatorname{cosec}\left(\theta + \frac{(m-1)\pi}{4}\right) \operatorname{cosec}\left(\theta + \frac{m\pi}{4}\right) = 4\sqrt{2}$$

is(are)

(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{6}$ (C) $\frac{\pi}{12}$ (D) $\frac{5\pi}{12}$

Answer



(A) (B) (C) (D)

SECTION – III

Matrix – Match Type

This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled p, q, r, s and t. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II**. The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:

If the correct matches are A – p, s and t; B – q and r; C – p and q; and D – s and t; then the correct darkening of bubbles will look like the following.

	p	q	r	s	t
A	●	○	○	●	●
B	○	●	●	○	○
C	●	●	○	○	○
D	○	○	○	●	●

29. Match the statements/expressions given in **Column I** with the values given in **Column II**.

Column I	**Column II**
(A) Root(s) of the equation $2\sin^2\theta + \sin^2 2\theta = 2$	(p) $\frac{\pi}{6}$
(B) Points of discontinuity of the function $f(x) = \left[\frac{6x}{\pi}\right]\cos\left[\frac{3x}{\pi}\right]$, where $[y]$ denotes the largest integer less than or equal to y	(q) $\frac{\pi}{4}$
(C) Volume of the parallelopiped with its edges represented by the vectors $\hat{i}+\hat{j}$, $\hat{i}+2\hat{j}$ and $\hat{i}+\hat{j}+\pi\hat{k}$	(r) $\frac{\pi}{3}$
(D) Angle between vectors $\vec{a}$ and $\vec{b}$ where $\vec{a}, \vec{b}$ and $\vec{c}$ are unit vectors satisfying $\vec{a}+\vec{b}+\sqrt{3}\vec{c}=\vec{0}$	(s) $\frac{\pi}{2}$
	(t) π

Answer

	p	q	r	s	t
A	○	●	○	●	○
B	●	○	●	●	●
C	○	○	○	○	●
D	○	○	●	○	○

30. Match the statements/expressions given in **Column I** with the values given in **Column II**.

Column I	**Column II**
(A) The number of solutions of the equation $x e^{\sin x} - \cos x = 0$ in the interval $\left(0, \frac{\pi}{2}\right)$	(p) 1
(B) Value(s) of k for which the planes $kx + 4y + z = 0$, $4x + ky + 2z = 0$ and $2x + 2y + z = 0$ intersect in a straight line	(q) 2
(C) Value(s) of k for which $\lvert x-1\rvert + \lvert x-2\rvert + \lvert x+1\rvert + \lvert x+2\rvert = 4k$ has integer solution(s)	(r) 3
(D) If $y' = y + 1$ and $y(0) = 1$, then value(s) of $y(\ln 2)$	(s) 4
	(t) 5

Answer

	p	q	r	s	t
A	●	○	○	○	○
B	○	●	○	●	○
C	○	●	●	●	●
D	○	○	●	○	○

SECTION – IV

Integer Answer Type

This section contains 8 questions. The answer to each of the questions is a single–digit integer, ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answers to question numbers X, Y, Z and W (say) are 6, 0, 9 and 2, respectively, then the correct darkening of bubbles will look like the following:

X	Y	Z	W
0	●	0	0
1	1	1	1
2	2	2	●
3	3	3	3
4	4	4	4
5	5	5	5
●	6	6	6
7	7	7	7
8	8	8	8
9	9	●	9

---

31. The maximum value of the function $f(x) = 2x^3 - 15x^2 + 36x - 48$ on the set $A = \{x \mid x^2 + 20 \le 9x\}$ is

Answer

0 1 2 3 4 5 6 **7** 8 9

32. Let (x, y, z) be points with integer coordinates satisfying the system of homogeneous equations:

$$3x - y - z = 0$$
$$-3x + z = 0$$
$$-3x + 2y + z = 0.$$

Then the number of such points for which $x^2 + y^2 + z^2 \leq 100$ is

Answer 0 1 2 3 4 5 6 **7** 8 9

33. Let ABC and ABC' be two non-congruent triangles with sides $AB = 4$, $AC = AC' = 2\sqrt{2}$ and angle $B = 30°$. The absolute value of the difference between the areas of these triangles is

Answer 0 1 2 3 **4** 5 6 7 8 9

34. Let $p(x)$ be a polynomial of degree 4 having extremum at $x = 1, 2$ and

$$\lim_{x \to 0} \left(1 + \frac{p(x)}{x^2}\right) = 2.$$

Then the value of $p(2)$ is

Answer **0** 1 2 3 4 5 6 7 8 9

35. Let $f : \mathbf{R} \to \mathbf{R}$ be a continuous function which satisfies

$$f(x) = \int_0^x f(t)dt.$$

Then the value of $f(\ln 5)$ is

Answer **0** 1 2 3 4 5 6 7 8 9

36. The centres of two circles C_1 and C_2 each of unit radius are at a distance of 6 units from each other. Let P be the mid point of the line segment joining the centres of C_1 and C_2 and C be a circle touching circles C_1 and C_2 externally. If a common tangent to C_1 and C passing through P is also a common tangent to C_2 and C, then the radius of the circle C is

Answer 0 1 2 3 4 5 6 7 **8** 9

37. The smallest value of k, for which both the roots of the equation

$$x^2 - 8kx + 16(k^2 - k + 1) = 0$$

are real, distinct and have values at least 4, is

Answer 0 1 **2** 3 4 5 6 7 8 9

38. If the function $f(x) = x^3 + e^{\frac{x}{2}}$ and $g(x) = f^{-1}(x)$, then the value of $g'(1)$ is

Answer 0 1 **2** 3 4 5 6 7 8 9